Dianna E. Ochoa Lynch

812.803.0086 | dianocly@mit.edu | www.linkedin.com/in/dianna-lynch
https://dianocly.github.io/Dianna-Ochoa-Lynch-portfolio/

Education

Massachusetts Institute of Technology (MIT) | Cambridge, MA *Expected May 2028*
Candidate for B.S. in Mechanical Engineering & Minor in Archaeology and Materials

- ***GPA** 4.9/5.0*
- **Relevant Coursework:** Mechanics and Materials II, Thermofluids Engineering II (In Progress), Design and Manufacturing II, Dynamics and Controls II (In Progress), Numerical Computation

Experience

Mechanical Engineering Intern - Boston Dynamics Waltham, MA
Atlas Actuation *Jun 2026–Aug 2026*

- Redesigned 3D-printed actuator components for injection molding in collaboration with injection molding suppliers, improving manufacturability and reducing projected production costs.
- Led material selection for actuator-internal plastics, balancing operating temperature, chemical compatibility, and injection molding requirements
- Root-caused assembly-stage actuator failures using failure-data tracking; wrote diagnostic guides used by internal build and repair teams and to support design improvements
- Co-designed a soldering fixture with a senior engineer to support manufacturing R&D for actuator technology
- Designed and fabricated actuator cross-sectional model for internal demonstrations and technical instruction

Robotics Researcher - Distributed Robotics Laboratory, CSAIL, MIT Cambridge, MA
Team: *Reporting to Prof. Daniela Rus and PhD Student Wil Norton* *Sep 2024–May 2026*
Project: *Tele-Hand Mk 1* — Anthropomorphic tendon-driven robotic hand for teleoperation and haptic feedback

- Co-designed a multi-DoF capstan tendon transmission to minimize backlash and maximize force fidelity for haptic rendering
- Designed and assembled a prototype 126:1 multi-stage capstan gearbox (design intent: ~1000°/s finger rotation), iterating helix geometry, reduction staging, and cable-tracking stability
- Designed finger skeletal structure and kinematics across four prototype generations for rapid iteration, integrating tendon routing and assembly constraints
- Validated tendon routing and stack-up with 3D-printed prototypes, resolving mechanical interferences before metal fabrication
- Led prototype manufacturing from CAD, drawings, and BOMs through tooling, machining, and assembly; wrote CNC programs for three prototype joints and verified bearing bores and flange lengths on a CMM, holding ±0.001 in
- Evaluated skeletal and transmission designs for anthropomorphism (ROM coverage, speed, and a 20 N grip-force requirement) to guide design trade-offs and test plans

Makerspace Lead & STEM Instructor Mollerussa, Spain
MIT Global Teaching Labs *Jan 2026–Feb 2026*

- Taught rapid prototyping and iterative design from sketch to CAD to prototype; maintained the makerspace's fabrication equipment and helped students use its tools

Mechatronics Engineer – Lead Designer and Systems Engineer Bloomington, IN
FIRST Robotics Competition Team 3494 & FIRST Tech Challenge Team 11329 *Aug 2021–May 2024*

- Directed the end-to-end design of 6 high-performance robots; ranked Top 10 twice out of ~8000 teams worldwide
- Recognized by General Motors for excellence in form-factor optimization, robustness, and serviceability—directly translating complex mechanical subsystems into a "production-ready" design
- Responsible for robot design under strict constraints and managing their fabrication, assembly, and programming

Technical Skills

- **Mechanical design:** precision mechanism design, tendon-driven actuation, gearbox design, weight optimization, DFM/DFA
- **Precision manufacturing:** GD&T (tolerance selection and fits), CNC programming (CAM), CMM inspection and metrology, manual mill and lathe, 3D printing, laser cutting
- **Analysis:** finite element analysis, MATLAB, kinematic modeling, root cause and failure analysis
- **Software:** SolidWorks, CATIA, Onshape, Inventor, Fusion 360, Arena, Fulcrum

Awards & Activities

MIT Design and Manufacturing I - Brabham BT46 "Fan Car" Prize – For a well designed and well executed robot in MIT's 2.007
Industrial Design Award (sponsored by General Motors) – 2024 FIRST Robotics Competition World Championship

- Recognized as 1 of 8 winners globally for excellence in form-factor optimization, aesthetics, robustness, and serviceability

2023 FRC Dean's List Finalist: Selected as 1 of 2 students from Indiana to represent the state at the FIRST World Championships

Activities: MIT Women's Ultimate Frisbee, Society of Women Engineers, Makerspace Mentor
Interests: Precision Mechanism Design, Manufacturing, Mechatronics, Soft Robotics and Compliant Mechanisms, Stained Glass